

WANTWOMBAN

Physical formalization of a
self-referential, 1-bit log-polar
thermochromic engine

FRAMEWORK ATTRIBUTION / AS SUPPLIED

Tom Klootwijk

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THE FORMAL OBJECT

A binary branching controller with explicit OTAN2 semantics, a log-polar pinion representation, a Klein-type seam identification, and a thermochromic material coupled through electrical power, heat transport, optical response, and measured feedback.

word → command → heat → colour → measurement → word

EDITION 1.0 / EXACT DEFINITIONS, PHYSICAL EQUATIONS, PROOFS AND REFERENCE KERNEL

Prepared from the supplied *Pasted markdown.md*. The original Markdown and executable reference kernel are embedded as PDF attachments.

Reading the formalization

WANTWOMBAN is the name requested for this edition. The source framework is attributed to Tom Klootwijk with the author designation printed on the cover. Names such as *OTAN2*, *Pinion Double Hadamard*, *PSI Fork Razor*, *mitosis*, *pinion nuclei*, *0-order notation*, and *thermochromatic paint* are retained. The conventional material-science adjective *thermochromic* is used alongside the source's *thermochromatic*. [S]

This edition has four kinds of statement. [D] denotes a mathematical or machine definition. [L] denotes an established physical law in its stated setting. [C] denotes a material, instrument, or implementation model with parameters. [P] denotes a consequence derived from the definitions or models. These labels distinguish what is chosen, what is measured, and what follows by calculation.

The attached source contains successive prompts and prior generated responses. Where those responses give incompatible layouts or equations, the present specification states the selected interpretation and records the difference in Section 13. An unchanged copy of the source remains embedded in the PDF. Added definitions and derivations belong to this edition; they are not presented as quotations from the source.

THE PHYSICAL CLOSURE

The engine's literal physical realization is the combination of a specified state transition, an energy-supplied actuator, a temperature-dependent coating, and a return measurement. A bit changes a control signal. The actuator transfers energy. The material changes its optical response. The sensor closes the self-referential loop. Each arrow has a defined variable and an equation.

Source data. The text supplies the conceptual architecture and example pseudocode. It supplies no measured coating composition, heat capacity, thermal conductivity, heater resistance, sensor calibration, or time-series measurements of biological cycles. Those quantities are explicit parameters below. Numerical values in the worked example are specified calculation inputs, not attributed measurements.

Implementation convention. The geometry/control kernel uses integer fields, comparisons, shifts, additions, and fixed tables. Its no-coordinate-squaring rule is retained. Continuous physical equations are written independently of the processor's instruction set; for example, Joule power is written as VI , and heat transport as a divergence of a flux.

DOCUMENT MAP

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1 The unified physical object

1.1 State, space and time

[D] Let the physical support be a substrate domain $\Omega \subset \mathbb{R}^3$, with coated surface $\Sigma \subset \partial\Omega$. Let t_n be successive controller update times, and let j identify a heater/sensor site. The complete state is

$$\mathcal{S}_n = (W_n, Q_n, \mathcal{L}, \mathbf{P}_n, T(\mathbf{x}, t_n), a(\mathbf{x}, t_n), \boldsymbol{\eta}_n). \quad (1.1)$$

Here W_n is one packed 64-bit geometry/control word; Q_n is the branching stack or queue; $\mathcal{L}$ is the lookup-table data; $\mathbf{P}_n$ contains the active pinion centres; T is absolute temperature; $a \in [0, 1]$ is the coloured-state fraction; and $\boldsymbol{\eta}_n$ holds sensor, hysteresis, optional phase-fraction and other explicitly retained memory. A word describes one active node. A collection of nodes is a collection of words, not extra information hidden inside one word.

[D] A node address is a finite bit string $\sigma \in \{0, 1\}^*$, with children $\sigma 0$ and $\sigma 1$. The source’s “BST” is used as a branching spatial tree. Calling it a *binary search tree* additionally requires an ordered key and a search invariant; the geometric construction alone requires neither. “Mitosis” denotes this two-child operation.

1.2 The causal closed-loop equations

[D,C] Let $G_n \in \{0, 1\}$ denote geometric occupancy and let $D_n \in \{0, 1\}$ denote measured optical darkness. Define

$$G_n = \mathcal{O}_g(W_n; \mathcal{L}), \quad (1.2)$$

$$D_n = \mathbf{1}\{R_n < R_*\}, \quad (1.3)$$

$$W_{n+1} = \delta(W_n, b_n, G_n, D_n), \quad (1.4)$$

$$u_n = 1 - (G_n \oplus \chi_n \oplus \kappa_n), \quad (1.5)$$

$$(T_{n+1}, a_{n+1}) = \Phi_{\Delta t_n}(T_n, a_n; u_n, \boldsymbol{\mu}), \quad (1.6)$$

where R_n is a calibrated reflectance measurement, R_* is its discrimination threshold, b_n is a branch bit, $\boldsymbol{\mu}$ contains physical parameters, and Φ is the physical evolution operator developed in Sections 8–9. The command u_n is held over $[t_n, t_{n+1})$. The next observation is made after that interval.

Equation (1.5) is an explicit *implementation choice*: heating is commanded for the light/clear target. The source’s binary semantic $1 = \text{on/occupied}$ is preserved for G_n , while $u_n = 1$ means heater on. These are separate channels. A coating with the opposite colour response uses a separately calibrated command decoder.

Proposition 1.1 (Causal self-reference). *Given an initial state, an input/branch sequence, finite lookup tables, and a uniquely solvable physical update, equations (1.2)–(1.6) determine a unique sequence of states.*

Proof. At step n , the current geometry and measured optical state are already known. They determine the next word and the held command. The physical update then determines the state at t_{n+1} . Induction determines every later state. No output is required to be its own instantaneous unknown input. $\square$

1.3 Units and the source’s numerical tokens

[D] The constants have separate roles:

$$\phi = \frac{1 + \sqrt{5}}{2}, \quad \phi = 1 + \phi^{-1}, \quad F_{n+2} = F_{n+1} + F_n, \quad F_0 = 0, F_1 = 1. \quad (1.7)$$

The ratio F_{n+1}/F_n approaches ϕ for $n \geq 1$. A full phase turn is 2π . The source’s $3.6 = 18/5$ and $36 = 10(3.6)$ are dimensionless numerical identities unless a unit assignment is declared. An applicable exact conversion

is $v_{\text{km/h}} = 3.6 v_{\text{m/s}}$. The supplied Gregorian date identifies 10 July 1990; the 36th anniversary is 10 July 2026. These are provenance and possible clock-origin inputs, not substitutions for an energy, length or temperature constant. [S]

Symbol	Meaning	Unit
$\rho, \theta, \Phi, \chi, \kappa$	Packed codes or dimensionless coordinates	1
$\mathbf{x}, \mathbf{P}_j, r_0, z_0$	Positions and length scales	m
$T, \Delta T$	Absolute temperature and temperature difference	K
C_j, G_{jk}, P_j	Heat capacity, conductance, supplied power	J/K, W/K, W
$k, q, \mathcal{H}$	Thermal conductivity, heat flux, enthalpy density	W/(m K), W/m ² , J/m ³
R, a, G_n, D_n, u_n	Reflectance, coloured fraction, binary channels	1

2 Clock, biological channels and 1-bit labels

2.1 Phase is a state; Fourier transformation is an operation

[D] The source’s “FT sweep” is retained as the name of the phase register. A periodic sweep is

$$p(t) = \text{frac}\left(\frac{t - t_0}{\tau_0}\right), \quad p_n = \frac{f_n}{32}, \quad f_n \in \{0, \dots, 31\}. \quad (2.1)$$

Its range is $[0, 1)$, and wrap closes the cycle. A Fourier transform, when actually used to analyse an input s , is the separate operation

$$\widehat{s}(\omega) = \int_{-\infty}^{\infty} s(t) e^{-i\omega t} dt. \quad (2.2)$$

Naming the sweep “FT” does not replace this integral. The reference controller updates $f_{n+1} = (f_n + 1 + D_n) \bmod 32$; the extra D_n is an explicit feedback choice in this edition.

2.2 Gestation, crop and menstrual modulation

[D,C] Keep the source channel labels g, c, m for gestation, crop and menstrual inputs. For an externally supplied positive interval function $\tau_j(t)$, define

$$\psi_j(t) = \psi_j(t_0) + 2\pi \int_{t_0}^t \frac{ds}{\tau_j(s)}. \quad (2.3)$$

The source proposes a logarithm of a product of trigonometric channels. Its finite-valued version is

$$x(t) = \varepsilon + (1 - \varepsilon) |\sin \psi_g(t) \cos \psi_c(t) \sin \psi_m(t)|, \quad 0 < \varepsilon < 1, \quad (2.4)$$

$$\gamma(t) = \frac{\ln x(t) - \ln \varepsilon}{-\ln \varepsilon}, \quad 0 \leq \gamma(t) \leq 1, \quad (2.5)$$

$$\gamma_J(t) = \text{clip}(\gamma(t) + \delta_J(2\eta_n - 1), 0, 1), \quad \eta_n \in \{0, 1\}. \quad (2.6)$$

Regularization by ε removes $\ln 0$; normalization supplies the nonnegative modulation range that the source’s pruning argument assumed. The unregularized logarithm of a product bounded by one is nonpositive, not nonnegative. The jitter bit is an explicitly generated or measured input; by itself it is not a probability distribution.

A lookup table may store the resulting samples so that the online loop contains no trigonometric or logarithmic evaluation. For measured cycles, store event timestamps and estimate each τ_j from the particular input series. The source provides no such series. Accordingly, its fertility-window and IVF references remain named input channels; no biological window or outcome is numerically inferred here.

2.3 The XX/XY and female/male wording

[D] The source uses several incompatible decoders for its chromosome-swap bit. To retain the explicit inverted naming in the originating prompt, define token $c = 0$ as “XX” and $c = 1$ as “XY”, and set

$$L_0 = \text{author's male-labelled channel}, \quad L_1 = \text{author's female-labelled channel}, \quad L(c, \chi) = L_{c \oplus \chi}. \quad (2.7)$$

Thus the base assignment is $XX \mapsto L_0$, $XY \mapsto L_1$, and $\chi = 1$ exchanges the two labels. This is a literal *symbol decoder*. No biological chromosome is modified by its XOR operation. The tokens “V” and “IVF” remain source annotations unless an additional input decoder is supplied. This preserves the author’s chosen language without inserting an unprovided biological mechanism.

[D] The two-bit selector is

$$\Xi = 00 : g, \quad 01 : c, \quad 10 : m, \quad 11 : \text{calibration/idle channel}.$$

The last assignment is an explicit completion of the previously unused selector value.

3 Exact 64-bit word and 0-order execution

3.1 Non-overlapping core fields

[D] Bits are numbered from 0 at the least significant bit. The authoritative layout is the following corrected version of the source’s original field table; it retains its widths and the later log-polar interpretation. [S]

Field	Bits	Range	Meaning
χ	1	63	Author-label/control polarity
ρ	11	62:52	Signed logarithmic-radius code
θ	11	51:41	Unsigned phase code, modulo 2048
z	11	40:30	Signed axial-position code
Φ	10	29:20	Unsigned phyllotaxis/tracking code
d	5	19:15	Tree depth, 0–31
f	5	14:10	Source’s FT/phase counter
ℓ	7	9:3	Polar-LUT address, 0–127
κ	1	2	Seam/orientation bookkeeping
Ξ	2	1:0	Cycle selector

The widths add to exactly 64. The source’s proposed thermal bits 8:5 overlap the LUT field. This edition leaves W intact and stores temperature in a separate register H_T . Geometric occupancy G_n and optical readback D_n are also separate one-bit channels; bit 0 of W remains part of Ξ .

[D] For width b and shift s , define $m_b = 2^b - 1$ and

$$\text{get}_{s,b}(W) = (W \gg s) \& m_b, \quad (3.1)$$

$$\text{put}_{s,b}(W, v) = (W \& \neg_{64}(m_b \ll s)) \mid ((v \& m_b) \ll s), \quad (3.2)$$

$$\text{signed}_b(v) = \begin{cases} v, & v < 2^{b-1}, \\ v - 2^b, & v \geq 2^{b-1}. \end{cases} \quad (3.3)$$

Here $\neg_{64}$ complements exactly 64 bits. The reference implementation rejects an out-of-range unsigned field value instead of silently masking it. The masks themselves appear in Appendix A.

Proposition 3.1 (Field isolation). *Replacing field (s, b) changes no bit outside its mask. Extracting and repacking every field reconstructs any unsigned 64-bit word exactly.*

Proof. The masks are pairwise disjoint and their union is all 64 positions. Replacement first clears only its own positions and then writes only those positions. Every other bit is unchanged, and the disjoint union gives exact reconstruction. $\square$

3.2 0-order means a defined evaluation language

[D] A 0-order expression is a left fold of its token stream:

$$E(a_0 \ o_1 \ a_1 \ \cdots \ o_m \ a_m) = o_m(\cdots o_2(o_1(a_0, a_1), a_2) \cdots, a_m). \quad (3.4)$$

Parenthesized subexpressions are evaluated first. All primitive operations act on explicitly sized integers. This is a language definition; it does not change C or Python operator precedence. The reference code therefore uses parentheses or one operation per assignment.

For example, the source’s intended low-three-bit parity is exactly

$$p_3(W) = (W \ \& \ 1) \oplus ((W \gg 1) \ \& \ 1) \oplus ((W \gg 2) \ \& \ 1). \quad (3.5)$$

Those three bits are Ξ_0, Ξ_1, κ , not three coordinate signs. At branch b , define $p_b = p_3(W) \oplus b$. The even branch shifts Φ left by one and masks to 10 bits; the odd branch shifts it right by one. Discarded bits are not preserved unless separately stored.

3.3 Exact toggles and signed inversion

[D] Toggling κ is $W' = W \oplus 4$. Adding 4 is not the same operation: when $\kappa = 1$, addition carries into bit 3. Numerical two’s-complement negation is $(-x) \bmod 2^b$; XOR with an all-ones field yields $2^b - 1 - x$, which represents $-x - 1$, not $-x$.

The signed 11-bit range is $[-1024, 1023]$. To allow an exactly reversible radial sign change, the seam profile uses the symmetric subset $[-1023, 1023]$ for ρ . The axial field may use the full signed range. An angular field wraps modulo 2048. A physical temperature does not wrap modulo 16: a display code is saturated after calibration.

4 OTAN2, logarithmic polar geometry and pinion nuclei

4.1 Exact OTAN2 definition

[D] Preserve the source’s “O” as the one-bit occupancy input. Define the explicitly grouped, 11-bit phase-mixing operator

$$\boxed{\text{OTAN2}_{G0}(r, O) = ((r \ \& \ 2047) \oplus (O \ll 10)) \ \& \ 2047.} \quad (4.1)$$

It takes a radius *code* and a Boolean state, not a pair of Cartesian coordinates. Consequently it is a defined phase generator rather than a computation of the geometric angle of an arbitrary point. For fixed O , it is a permutation of the 2048 radius codes. When $O = 1$, it toggles the most significant phase bit, equivalently adding a half turn modulo 2048.

The source also gives an unparenthesized expression and asks for literal left-to-right execution. That literal profile is different:

$$\text{OTAN2}_{Z0}(r, O) = (((r \ \& \ 2047) \oplus O) \ll 10) \ \& \ 2047. \quad (4.2)$$

Its image is exactly $\{0, 1024\}$: only the lowest bit before the shift survives. Both profiles are formalized. The executable kernel uses $G0$, with the grouping made explicit; $Z0$ remains available as `otan2_literal`. This is a recorded disambiguation, not an assumption that the two formulas are equivalent.

4.2 From codes to physical positions

[D,C] A pinion nucleus is a local coordinate centre $\mathbf{P}_\sigma$ and an address in the branching controller. Its physical realization may be a heater site, a sensor site, or an indexed location. The word “nucleus” here does not assign nuclear forces or atomic composition.

Choose a reference length $r_0 > 0$, a log-radius increment $\Delta_\rho > 0$, an axial increment $\Delta_z > 0$, and an orthonormal local frame $(\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3)$. The physical realization map is

$$\varrho = \Delta_\rho \text{signed}_{11}(\rho), \quad \vartheta = 2\pi\theta/2048, \quad (4.3)$$

$$r = r_0 \exp((\ln 2)\varrho), \quad z_{\text{phys}} = \Delta_z \text{signed}_{11}(z), \quad (4.4)$$

$$\mathbf{x} = \mathbf{P}_\sigma + r \cos \vartheta \mathbf{e}_1 + r \sin \vartheta \mathbf{e}_2 + z_{\text{phys}} \mathbf{e}_3. \quad (4.5)$$

The logarithm applies to the dimensionless ratio r/r_0 . The chart represents $r > 0$; the centre $r = 0$ is a separately tagged apex, not a finite log-radius value. The controller may obtain every required exponential and direction from a precomputed table.

For nearest quantization, the log-radius code error is at most $\Delta_\rho/2$, so the relative radius error is at most $2^{\Delta_\rho/2} - 1$. The nearest phase error is at most $\pi/2048$. These are encoding errors, not changes in physical dimensionality. More nested centres require more node addresses or successive evaluations.

4.3 Kinematic calculus

[P] With a fixed local frame and a continuously interpolated trajectory, define $\mathbf{e}_r = \cos \vartheta \mathbf{e}_1 + \sin \vartheta \mathbf{e}_2$ and $\mathbf{e}_\vartheta = -\sin \vartheta \mathbf{e}_1 + \cos \vartheta \mathbf{e}_2$. Differentiating (4.5) gives

$$\dot{\mathbf{x}} = \dot{\mathbf{P}}_\sigma + r(\ln 2)\dot{\varrho} \mathbf{e}_r + r\dot{\vartheta} \mathbf{e}_\vartheta + \dot{z}_{\text{phys}} \mathbf{e}_3. \quad (4.6)$$

This supplies the requested kinematic relation without coordinate squaring. If a moving physical actuator follows the trajectory, its mechanics additionally obey $m\dot{\mathbf{v}} = \mathbf{F}_{\text{actuator}} + \mathbf{F}_{\text{external}}$ and require mechanical power $\mathbf{F} \cdot \mathbf{v}$. A change in an address alone is not physical translation of matter.

4.4 Golden-ratio leaf increments

[D] Preserve the source increment $\alpha = 2\pi/\phi$. Its complementary golden-angle convention is $2\pi - \alpha = 2\pi/\phi^2$. A 2048-step phase ring uses

$$a_\phi = \text{rnd}(2048/\phi) = 1266, \quad 2048 - a_\phi = 782. \quad (4.7)$$

The reference loop adds +1266 at even parity and −1266 at odd parity. Exact irrational rotation and finite cyclic rotation are separate versions: the latter is explicitly quantized. A pure phase accumulator with increment 1266 has period $2048/\text{gcd}(2048, 1266) = 1024$.

5 Pinion Double Hadamard: matrix and arithmetic

5.1 Exact real-valued operator

[D] Retain the source’s normalized tensor Hadamard construction, with all entries specified:

$$H_4 = H_2 \otimes H_2 = \frac{1}{2} \begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & -1 & 1 & -1 \\ 1 & 1 & -1 & -1 \\ 1 & -1 & -1 & 1 \end{pmatrix}, \quad D_\phi = \text{diag}(\phi, \phi^{-1}, 1, -1). \quad (5.1)$$

The source's elementwise pinion multiplication is precisely

$$A_\phi \mathbf{v} = D_\phi H_4 \mathbf{v} = (H_4 \mathbf{v}) \odot (\phi, \phi^{-1}, 1, -1). \quad (5.2)$$

The coordinates in $\mathbf{v}$ are normalized dimensionless components. Dimensional positions, phase angles and log-radius codes are first converted to a declared coordinate chart; bare metres are not added to radians.

Proposition 5.1 (The actual invariants). $H_4^T H_4 = I$, $H_4 H_4 = I$, $\det H_4 = 1$, and $\det A_\phi = -1$. Therefore A_ϕ preserves four-dimensional volume magnitude. It does not in general preserve lengths, areas, or individual component magnitudes.

Proof. Distinct sign rows have zero dot product and each row has four unit products before the factor $1/2$ is applied twice. This gives orthogonality and the involution identity. The tensor determinant gives $\det H_4 = 1$, while $\det D_\phi = \phi \phi^{-1} (1)(-1) = -1$. Since $D_\phi^T D_\phi \neq I$, the combined operator is not orthogonal. Its singular values are $\phi, \phi^{-1}, 1, 1$. $\square$

Volume preservation here is a mathematical Jacobian statement. It is not conservation of physical mass, thermal energy, or “informational mass”. Neither branching nor discarding a branch becomes reversible merely because this real matrix is invertible.

5.2 Integer butterfly and fixed-point pinion

[D] Write $B = 2H_4$. For $\mathbf{v} = (x, y, z, f)$,

$$s_0 = x + y + z + f, \quad s_1 = x - y + z - f, \quad (5.3)$$

$$s_2 = x + y - z - f, \quad s_3 = x - y - z + f. \quad (5.4)$$

Then $BB = 4I$ and $\det B = 16$. The normalized entries are $\pm 1/2$, not ± 1 . The butterfly uses additions and subtractions; normalization still has to be implemented.

For a fixed-point scale $Q = 65536$, set

$$a = 106039 = \text{rnd}(Q\phi), \quad b = 40503 = \text{rnd}(Q/\phi). \quad (5.5)$$

The reference integer map is

$$\hat{A}(\mathbf{v}) = \left(\text{rnd} \frac{as_0}{2Q}, \quad \text{rnd} \frac{bs_1}{2Q}, \quad \text{rnd} \frac{s_2}{2}, \quad \text{rnd} \frac{-s_3}{2} \right), \quad (5.6)$$

where nearest rounding resolves exact ties away from zero. Multiplication by a or b can be evaluated exactly as a sum of shifted copies of an integer. The irrational factors themselves are approximated by the specified rational coefficients.

[P] For the first component, coefficient quantization contributes no more than $|s_0|/(4Q)$ code units; final rounding contributes at most $1/2$ code unit. The second component has the analogous bound. Overflow is avoided with wider intermediate integers; only the final field is clipped or wrapped according to its declared type.

The fixed-point implementation does not inherit exact volume preservation. As a simple loss-of-invertibility example, using equal-width modular fields gives

$$B(1024, 1024, 0, 0)^T = (2048, 0, 2048, 0)^T \equiv \mathbf{0} \pmod{2048}.$$

A reversible implementation retains enough intermediate width and any discarded remainders. The compact reference profile instead specifies its rounding, saturation and wrap explicitly.

6 Bifurcation, parity and the Klein closure

6.1 A complete node transition

[D] The following definition completes the source's algorithm. Source-derived components are the Hadamard butterfly, parity shifts, phase/LUT fields and terminal feedback. The rational coefficients, clipping policy, unit radial step, grouped OTAN2 and optical phase increment are explicit choices of this edition.

Given W_n, b_n, D_n , first calculate G_n by Section 7 and $p = p_3(W_n) \oplus b_n$. Apply the parity shift to Φ_n . Decode $(\rho_n, \theta_n, z_n, \Phi_n)$ to signed code representatives, then apply (5.6) to obtain (r_H, a_H, z_H, f_H) . Define

$$r' = \text{clip}(r_H + 2b_n - 1, -1023, 1023), \quad (6.1)$$

$$\theta' = [a_H + \text{OTAN2}_{G0}(r' \bmod 2048, G_n) + (1 - 2p)1266] \bmod 2048, \quad (6.2)$$

$$z' = \text{clip}(z_H, -1024, 1023), \quad \Phi' = f_H \bmod 1024, \quad (6.3)$$

$$f' = (f_n + 1 + D_n) \bmod 32. \quad (6.4)$$

Store the sampled LUT index in ℓ' . Retain χ and Ξ . If $d_n < 31$, set $d' = d_n + 1$ and retain κ . At $d_n = 31$, apply the seam defined below and reset $d' = 0$. The same parent can be evaluated with $b = 0$ and $b = 1$ to produce two independent child words. Updating one child must not mutate the input used for the other.

6.2 The actual Klein identification

[D] A Klein bottle is defined by a quotient, not by a chosen eigenvalue. Use the rectangle with

$$(u, 0) \sim (-u \bmod 1, 1), \quad (0, v) \sim (1, v). \quad (6.5)$$

This is the orientation-reversing edge identification described in Hatcher's topology notes. Its local transition reverses one tangent direction. An ordinary three-dimensional depiction has a self-intersection; a control implementation can instead wire or address the paired edges according to (6.5). [10]

For the packed chart, take $u = (\theta - 512)/2048 \bmod 1$ and $v = d/32$. A terminal crossing therefore uses the shifted reflection

$$\theta^K = (1024 - \theta') \bmod 2048, \quad \kappa^K = \kappa \oplus 1. \quad (6.6)$$

The reference profile separately includes the source's reciprocal log-radius action, centred at zero:

$$r^K = -r', \quad r_{\text{phys}}^K = \frac{r_0 r_0}{r_{\text{phys}}}. \quad (6.7)$$

Equation (6.6) defines the Klein base identification. Equation (6.7) is an additional involution on the radius attached to that base; it is not needed to define the Klein surface. Switching κ may select a paired centre $\mathbf{P}_0 \leftrightarrow \mathbf{P}_1$. The full state, which also contains radius, axial position and material variables, is richer than the two-dimensional Klein base.

Proposition 6.1 (Seam involution). *Applying the phase reflection, radial inversion and κ toggle twice recovers all three input fields on the symmetric radial range.*

Proof. $1024 - (1024 - \theta) \equiv \theta \pmod{2048}$, $-(-r) = r$, and $(\kappa \oplus 1) \oplus 1 = \kappa$. □

A half-turn alone, $\theta \mapsto \theta + 1024$, is not the reflection in (6.5). The source's half-turn is retained inside the shifted-reflection expression but supplemented by the required sign reversal. Likewise, the local derivative has a -1 eigenline in the reversed tangent direction; setting an unrelated global eigenvalue to -1 is not a construction of a quotient space.

6.3 Self-reference and the eigenvector-axis wording

[D,P] If a branch path has affine map $\mathbf{v} \mapsto A_\sigma \mathbf{v} + \mathbf{t}_\sigma$ and terminal linear seam K , a fixed root satisfies

$$(I - KA_\sigma)\mathbf{v}_* = K\mathbf{t}_\sigma. \quad (6.8)$$

There is a unique affine fixed point when this matrix is invertible. In the homogeneous case, a fixed nonzero vector satisfies $KA_\sigma \mathbf{v}_* = \mathbf{v}_*$. Iteration converges to the affine fixed point when the spectral radius of KA_σ is less than one. None of these properties follows solely from naming an axis “Klein”.

6.4 Non-chirality as an explicit symmetry

[D] Binary labels 0 and 1 have no inherent geometric handedness. Nonorientability and mirror symmetry are distinct properties. A scalar occupancy on the quotient must agree on identified edges: $G(u, 0) = G(-u, 1)$. A complemented value across the seam instead represents an orientation-dependent channel on the double cover.

A practical mirror-symmetric occupancy is $G_{\text{sym}}(\mathbf{x}) = G(\mathbf{x}) \vee G(M\mathbf{x})$ for a reflection M with $MM = I$. Then $G_{\text{sym}}(M\mathbf{x}) = G_{\text{sym}}(\mathbf{x})$ exactly. The unconstrained reference LUT need not have this symmetry; it can be imposed when compiling a symmetric specimen.

7 Geometry, polar LUTs and the PSI Fork Razor

7.1 Occupancy and signed distance are separate objects

[D] Preserve $O = 1$ for occupied/on and $O = 0$ for unoccupied/off. For a region A ,

$$O_A(\mathbf{x}) = \mathbf{1}\{\mathbf{x} \in A\}. \quad (7.1)$$

A genuine signed distance also carries a magnitude: it is minus the distance to the boundary inside A and plus that distance outside. Thresholding a signed distance produces occupancy; the one-bit result is not itself a signed distance. On a nonorientable base, occupancy is defined chartwise with the seam agreement just given, rather than by a globally oriented normal.

The source’s labels “sphere”, “cone”, “pyramid”, “apex” and “leaves” are made geometrically explicit without runtime coordinate squaring:

Primitive	Exact definition or declared implementation
Apex	The tagged point $\mathbf{P}_\sigma$. An L^1 neighbourhood is $ x + y + z \leq \epsilon$, not a point.
Diamond-base pyramid	$0 \leq z \leq h$ and $ x + y \leq s(h - z)$, with base at $z = 0$ and apex at $z = h$.
Circle / sphere	All support inequalities $\mathbf{n} \cdot (\mathbf{x} - \mathbf{P}) \leq R$ for unit directions $\mathbf{n}$ in two / three dimensions.
Circular cone	$0 \leq z \leq h$ and all planar support inequalities $\mathbf{n} \cdot (x, y) \leq s(h - z)$.
Leaf	A specified star-shaped polar boundary $r \leq r_{\max}(\vartheta, z)$, translated to its pinion centre and placed at its leaf phase.

The source expression $|x| + |y| - |z| \leq 0$ instead describes an unbounded double pyramidal cone. A finite list of directions gives a polygon/polyhedron approximation to a circle/sphere. If its directions cover every true direction within angle $\delta < \pi/2$, its outer radial error is bounded by $R(\sec \delta - 1)$: along any ray, a listed normal has dot product at least $\cos \delta$ with that ray.

7.2 Logarithmic boundary table

[D,C] For a star-shaped section with a positive boundary radius,

$$\lambda(\vartheta, z) = \log_2 \frac{r_{\max}(\vartheta, z)}{r_0}, \quad G = \mathbf{1}\{\varrho \leq \lambda(\vartheta, z)\}. \quad (7.2)$$

For the 128-entry reference table, store integer log-radius boundaries L_j . Define

$$j_n = ((\theta_n \gg 4) + f_n + 13\Xi_n) \bmod 128, \quad (7.3)$$

$$\eta_n = (\Phi_n \& 1) \oplus \kappa_n, \quad (7.4)$$

$$G_n = \mathbf{1}\{\text{signed}_{11}(\rho_n) \leq L_{j_n} + 2\eta_n - 1\}. \quad (7.5)$$

The supplied reference table is exactly $L_j = 128$ for all $j = 0, \dots, 127$. It is a constant radial boundary with one-code jitter. It is not labelled a full three-dimensional sphere: a sphere needs an axial-dependent table or support tests. The coefficient 13 is a specified address-mixing choice, not an inferred biological frequency. All 128 entries are thus reproducible without an unspecified data array.

7.3 Tilt-shifted slit and conservative pruning

[D] Let the observation slit be the slab

$$\mathcal{P} = \{\mathbf{x} : |\mathbf{n} \cdot \mathbf{x} - c| \leq w/2\}, \quad (7.6)$$

where $\mathbf{n}$ is a unit normal, c locates the slit, and $w \geq 0$ is its width. Its tilt is in $\mathbf{n}$. A pinion-centred bound has centre $\mathbf{P}_\sigma$ and radius R_σ enclosing *every descendant primitive to be considered in the current traversal epoch*.

Proposition 7.1 (PSI Fork Razor). *If*

$$|\mathbf{n} \cdot \mathbf{P}_\sigma - c| > w/2 + R_\sigma, \quad (7.7)$$

then none of those descendant primitives intersects the slit.

Proof. For any enclosed point, $|\mathbf{n} \cdot (\mathbf{x} - \mathbf{P}_\sigma)| \leq R_\sigma$. The reverse triangle inequality gives $|\mathbf{n} \cdot \mathbf{x} - c| \geq |\mathbf{n} \cdot \mathbf{P}_\sigma - c| - R_\sigma > w/2$. $\square$

For a finite tree, compute a conservative subtree hull by taking the hull of the node's geometry and both child hulls. For a separately chosen contractive construction with child-centre increments at depth d bounded by $\ell_0\beta^d$, primitive radii at most $R_0\beta^d$, and $0 < \beta < 1$, one valid envelope is

$$R_\sigma = R_0\beta^d + \frac{\ell_0\beta^d}{1 - \beta}. \quad (7.8)$$

The geometric series proves this bound. These are sufficient geometric hypotheses, not consequences of Hadamard mixing. Source rules such as $|x| > 32 - d$ have no dimensional or geometric justification without an equivalent reachability bound.

Geometric exclusion permits skipping pure geometry evaluation, not discarding state transitions that affect later feedback. Off-slit heaters can also influence visible sites by thermal conduction. Such transitions and coupled heat sources must be retained or replaced by a proven equivalent summary.

Terminal feedback starts a new epoch. Pruning a current subtree does not prune the future root into which it feeds. An infinite-time claim requires a separate invariant reachable set containing that root. Output uses an ordered list of slit samples or physical sites, without polygon rasterization, raymarching or raytracing. A finite binary output is still sampled data; culling can reduce work but has no guaranteed constant-time bound for arbitrary scenes.

8 Physical laws of the thermochromic realization

8.1 Energy balance and electrical transduction

[L] For a closed material system, use the first-law sign convention

$$\frac{dU}{dt} = P_{\text{electrical}} + P_{\text{mechanical}} + \dot{Q}_{\text{in}} - \dot{Q}_{\text{out}}. \quad (8.1)$$

Positive power is supplied to the system. This is conservation of energy; the bit word and the topology select the control schedule, not a replacement conserved quantity. [5]

[L,C] In a resistive, quasistatic conductor with negligible induction and charge accumulation, the electrical coupling is

$$\mathbf{E} = -\nabla V, \quad \mathbf{J} = \sigma_e(T)\mathbf{E}, \quad \nabla \cdot \mathbf{J} = 0, \quad q_J = \mathbf{J} \cdot \mathbf{E}. \quad (8.2)$$

For a heater site with a calibrated on-state voltage and current,

$$P_j(t) = u_j(t)V_j(t)I_j(t), \quad u_j \in \{0, 1\}. \quad (8.3)$$

The product is actual electrical power in watts. Current conservation, Ohm's law and heat generation are the physical connection implemented by a Joule-heating model. [8]

A software bit flip does not itself place heat at the coordinate encoded by the word. A physical address decoder and an actuator deliver the supplied power to that site. Processor heat is included separately through a measured spatial coupling q_{logic} , rather than assumed to appear at every pinion centre.

8.2 Heat transport in the substrate and coating

[L,C] Let $\mathcal{H}(T, \xi)$ be enthalpy per unit volume, where ξ is a material phase fraction distinct from selector Ξ . Use

$$\frac{\partial \mathcal{H}}{\partial t} + \nabla \cdot \mathbf{q} = q_J + q_{\text{logic}} + q_{\text{optical}}, \quad (8.4)$$

$$\mathbf{q} = -k(T, a)\nabla T, \quad (8.5)$$

$$\mathcal{H}(T, \xi) = \int_{T_{\text{ref}}}^T c_v(\tau) d\tau + L_v \xi. \quad (8.6)$$

Here c_v denotes volumetric sensible heat capacity (not a mass-specific constant-volume symbol), and L_v is latent enthalpy per unit volume when a phase-change model is used. In a fixed phase with constant properties, $c_v = \rho_m c_p$, and equation (8.4) reduces to $\rho_m c_p \partial_t T = \nabla \cdot (k \nabla T) + q_{\text{sources}}$. Fourier heat flux and this balance are the standard conductive solid model. [7]

[C] On an exposed surface with outward normal $\mathbf{n}$, ambient temperature T_a , and radiative surroundings T_r ,

$$\mathbf{q} \cdot \mathbf{n} = h_c(T - T_a) + \epsilon_r \sigma_{\text{SB}} [\mathcal{R}(T) - \mathcal{R}(T_r)], \quad \mathcal{R}(T) = T T T T. \quad (8.7)$$

This writes the fourth-power radiation law as repeated products. It assumes a diffuse gray surface facing effectively uniform surroundings in an index-one medium. For ideal internal contact, temperature and normal heat flux are continuous; a measured contact resistance instead produces a temperature jump proportional to interface heat flux. [7]

The constant $\sigma_{\text{SB}} = 5.670374419 \dots \times 10^{-8} \text{ W m}^{-2} \text{ K}^{-4}$ is the Stefan–Boltzmann constant. Absolute temperatures, not Celsius numerical values, enter this radiation expression. [4]

8.3 Entropy balance and the true meaning of a reset

[L,C] For local thermal equilibrium without mass transport, the conductive and electrical contributions to entropy production are

$$\dot{s}_{\text{prod}} = \frac{k \nabla T \cdot \nabla T}{T T} + \frac{\mathbf{J} \cdot \mathbf{E}}{T} + \dot{s}_{\text{material}} \geq 0. \quad (8.8)$$

Material kinetics must be chosen with nonnegative material entropy production. At the device-plus-environment level, the second law is $\dot{S}_{\text{total}} \geq 0$. The choice of bit-operation precedence does not determine that sign; the physical transport and dissipation do. Heat flows passively from hotter to colder parts under these conductances. [6, 7, 8]

[L] For erasure of one initially unknown, equiprobable bit in a symmetric memory returned to its standard state while coupled to a bath at T , the ideal average heat bound is

$$\langle Q_{\text{bath}} \rangle \geq k_B T \ln 2. \quad (8.9)$$

For h bits of logical information lost under that model, the bound is $h k_B T \ln 2$. Logical uncertainty must account for retained side information. It is not the Hamming distance between two observed words. NOT, XOR with a known constant, or a reversible permutation can change bits without logical erasure. These distinctions follow from Landauer's erasure argument and Bennett's reversible computation construction. [1, 2]

The exact SI Boltzmann constant is $k_B = 1.380649 \times 10^{-23}$ J/K. At 300 K, the bound per erased bit is approximately 2.87098×10^{-21} J. It is a bound on an average thermodynamic process, not a mandatory heat quantum emitted by every shift. [3, 1]

8.4 Thermal zero and algorithmic zero

[L,D] The third law concerns low-temperature thermodynamics; its unattainability form concerns the resources required to reach absolute zero. It does not identify the all-zero digital word with zero temperature. The latter is a register state. A device can hold $W = 0$ at any temperature within its operating model. An absolute-zero reset is therefore not part of δ . [11]

THE SOURCE'S "UNIVERSAL PRAGMATIC LAWS" IN PHYSICAL FORM

Energy is conserved by (8.1). Entropy production is nonnegative for the passive transport and dissipative material model. Erasure is governed by (8.9) in its stated memory model. The coating responds through its material law. Topological parity changes an address or command; it does not independently create a heat source, a heat sink, or a heat pump.

9 Thermochromic material, optical state and feedback

9.1 Colour is a constitutive response

[C] Let $a = 1$ represent the calibrated coloured endpoint and $a = 0$ the cleared or alternate-colour endpoint. The coating's spectral reflectance is a measured function

$$R_\lambda = \mathcal{R}_\lambda(T, a, \xi; \text{composition, thickness, substrate}). \quad (9.1)$$

For a first fitted model, one may use the endpoint mixture $R_\lambda = a R_{\lambda, \text{col}} + (1 - a) R_{\lambda, \text{clear}}$. This interpolation is a chosen model, not a universal optical mixing law.

Experimental work on leuco-dye composites shows that colour change and phase change can have different temperature dependence and thermal history, particularly during cooling. This supports retaining separate colour state a , phase state ξ , and heating/cooling history rather than one universal threshold or an assumption of complete transparency. [9]

9.2 Two useful levels of material model

[C] The ideal binary hysteresis model uses two measured temperatures $T_{\downarrow} < T_{\uparrow}$:

$$D_{n+1} = \begin{cases} 0, & T_{n+1} \geq T_{\uparrow}, \\ 1, & T_{n+1} \leq T_{\downarrow}, \\ D_n, & T_{\downarrow} < T_{n+1} < T_{\uparrow}. \end{cases} \quad (9.2)$$

This is a calibrated darkness convention for a coating that lightens on heating. The retained previous state is necessary in the middle interval. It is an idealization of a measured hysteresis curve, not a statement that every thermochromic pigment is fully opaque below one temperature and fully transparent above it.

A smooth dynamic model uses

$$\tau_a(T)\dot{a} = a_{\text{eq}}(T, h) - a, \quad a_{\text{eq}}(T, h) = \frac{1}{1 + \exp((T - T_c(h))/\Delta_c)}, \quad \Delta_c > 0, \quad (9.3)$$

with branch/history state h . Initial $a \in [0, 1]$ remains in that interval: at $a = 0$ the derivative is nonnegative and at $a = 1$ it is nonpositive. Fit τ_a , $T_c(h)$ and Δ_c to the particular coating. Where latent heat is independently measured, fit ξ and L_v separately in the heat balance. A complete equilibrium phase submodel can take $\xi(T) = [1 + \exp(-(T - T_{\xi})/\Delta_{\xi})]^{-1}$ with $\Delta_{\xi} > 0$ and $L_v \geq 0$. It gives a positive effective heat capacity $c_v + L_v d\xi/dT$ when $c_v > 0$. A dynamic phase fraction instead needs its own kinetic equation and retained state. The one-site reference calculation uses $L_v = 0$.

9.3 Measured 1-bit optical output

[C,D] For illumination E_{λ} and detector weighting w_{λ} , a normalized observable is

$$R_n = \frac{\int w_{\lambda} E_{\lambda} R_{\lambda}(t_n) d\lambda}{\int w_{\lambda} E_{\lambda} d\lambda}, \quad D_n = \mathbf{1}\{R_n < R_*\}. \quad (9.4)$$

A real implementation measures this quantity or an equivalent calibrated sensor signal. The source's proposed expression $G \oplus$ transparency is a valid *logical* channel, but it equals a physical optical output only when the optical stack and readout implement that truth table. Ordinary colour mixing is not defined by XOR.

A hard binary threshold is a classical readout choice. Microscopic spectroscopy or a calibrated molecular free-energy model would be additional information for a microscopic description; calling the coating a “quantum threshold filter” does not supply those data.

9.4 Temperature register and fail-safe control

[D,C] A 4-bit display code, when needed, is

$$H_4(T) = \text{clip}\left(\text{rnd}\left[15 \frac{T - T_{\min}}{T_{\max} - T_{\min}}\right], 0, 15\right), \quad T_{\max} > T_{\min}. \quad (9.5)$$

It is outside W . For the transport profile, the temperature register is instead a 16-bit centikelvin code, $H_T = \text{clip}(\text{rnd}(100T), 0, 65535)$, accompanied by an overrange flag. Neither code is the material temperature itself.

The physical driver applies a hardware-rated current limit and a measured temperature cutout:

$$u_{\text{actual}} = u_n \mathbf{1}\{T < T_{\text{cut}}\} \mathbf{1}\{\text{sensor and driver healthy}\}. \quad (9.6)$$

T_{cut} is selected from the coating, substrate, adhesive and driver ratings; the source supplies none of those ratings. A controller reset or a detected sensor fault commands the heater off. This makes the physical actuation rule complete rather than leaving a logical loop with an unspecified energy source.

10 Thermal network and worked physical example

10.1 A network of pinion sites

[C] When each site can be represented by one temperature, use a lumped network:

$$C_j \dot{T}_j = P_j - G_{j\infty}(T_j - T_a) - \sum_{k \neq j} G_{jk}(T_j - T_k), \quad (10.1)$$

with $C_j > 0$, $G_{j\infty} \geq 0$ and symmetric $G_{jk} = G_{kj} \geq 0$. These coefficients are obtained from geometry, a spatial heat model, or fitted heating/cooling data. Internal spatial uniformity is part of this model; a resolved version uses (8.4) instead. [7]

Proposition 10.1 (Internal heat exchange conserves energy). *The pairwise exchange terms in (10.1) cancel in the sum over all sites.*

Proof. The terms for j, k are $-G_{jk}(T_j - T_k)$ and $-G_{kj}(T_k - T_j)$. Symmetry makes their sum zero. Thus $\sum_j C_j \dot{T}_j = \sum_j P_j - \sum_j G_{j\infty}(T_j - T_a)$. $\square$

[P] Pairwise entropy production is

$$G_{jk}(T_j - T_k) \left(\frac{1}{T_k} - \frac{1}{T_j} \right) \geq 0 \quad (10.2)$$

for positive temperatures. Electrical work converted to heat contributes $P_j/T_j \geq 0$. Relabelling j by a seam permutation preserves these balances; changing κ is not changing the sign of a passive conductance.

10.2 Specified one-site calculation

[C] The following values define an illustrative specimen calculation. They are not measurements of the supplied framework or a particular paint.

Input	Value	Role
C	0.020 J/K	Effective thermal capacity
G	0.005 W/K	Aggregate conductance to ambient
$P_{\max}$	0.100 W	Heater on-state power
T_a	293.15 K (20 °C)	Ambient and initial temperature
$T_{\uparrow}$	304.15 K (31 °C)	Illustrative clear-on-heating threshold
$T_{\downarrow}$	301.15 K (28 °C)	Illustrative colour-on-cooling threshold

For held command u , the exact model solution is

$$T(t + \Delta t) = T_a + \frac{P_{\max} u}{G} + \left(T(t) - T_a - \frac{P_{\max} u}{G} \right) e^{-G\Delta t/C}. \quad (10.3)$$

The time constant is $\tau = C/G = 4.000$ s. The on-state equilibrium is 313.15 K (40 °C), while the off-state equilibrium is ambient. The modeled first passage from ambient to $T_{\uparrow}$ is

$$t_{\text{first}} = -\tau \ln \left(1 - \frac{G(T_{\uparrow} - T_a)}{P_{\max}} \right) = 3.19403 \text{ s}. \quad (10.4)$$

Energy supplied over this first heating interval is 0.319403 J. Stored sensible energy is $C(T_{\uparrow} - T_a) = 0.220000$ J. The remaining 0.099403 J is modeled heat loss. These three numbers satisfy the first law exactly up to rounding.

10.3 A physically self-referential thermal oscillator

[D,P] As a separate thermal-loop test, let the optical hysteresis bit command its own next heater interval: heater on while the material is coloured, heater off after it clears. This isolates material feedback from the geometry kernel. From the lower threshold to the upper threshold, and back,

$$t_{\text{on}} = \tau \ln \frac{P_{\text{max}}/G - (T_{\downarrow} - T_a)}{P_{\text{max}}/G - (T_{\uparrow} - T_a)} = 1.15073 \text{ s}, \quad (10.5)$$

$$t_{\text{off}} = \tau \ln \frac{T_{\uparrow} - T_a}{T_{\downarrow} - T_a} = 1.27381 \text{ s}. \quad (10.6)$$

The ideal continuous-event period is 2.42454 s, with duty fraction 0.474617 and mean heater power 0.0474617 W. Sampled switching adds timing quantization and sensor/material delay. The full WANTWOMBAN geometry-driven system uses (1.5); this oscillator is a distinct, fully specified physical closure example.

10.4 Information erasure is not the heater budget

[P] At 300 K, even a hypothetical rate of 10^6 complete erasures per second of 64 independently unknown bits has an ideal Landauer bound of only

$$64(10^6)(2.87098 \times 10^{-21}) = 1.83743 \times 10^{-13} \text{ W}.$$

The specified 0.100 W heater power is about 5.44×10^{11} times that bound. This calculation uses the erasure model of Section 8; it is not an estimate of the actual processor power. The heater is powered by the electrical supply. Counting bit transitions cannot substitute for measuring voltage, current and thermal coupling.

11 Physical output and reproducible transport

11.1 Non-rasterized physical addressing

[D,C] The reference physical arrangement is a coated substrate with individually addressable resistive sites, measured temperature, and optical readback. An ordered traversal of accepted tree nodes selects sites along the slit. Its address list may be linear, branched, or paired at the Klein seam. It need not be a two-dimensional raster.

The heat equation still governs spreading between physical sites. Increasing the count of symbolic leaves does not prevent thermal cross-talk. Resolve this with the spatial model or fit the network conductances G_{jk} . An addressing seam is implemented by a mapping between site identifiers; a drawn three-dimensional Klein self-intersection does not by itself create the corresponding wiring or material contact.

11.2 WMB1 packet specification

[D] The source's line-packet labels and byte offsets disagree. The following transport completion is internally consistent and carries the original 64-bit word unchanged. Integers are unsigned and big-endian. There is one active-site temperature sample per packet.

Byte offset	Width	Field
0–3	4 bytes	ASCII magic WMB1.
4	1 byte	Flags: bit 0 χ , bit 1 κ , bit 2 D , bit 3 u ; bit 4 temperature overrange; bit 5 thermal/fault cutout; bits 6–7 zero.
5	1 byte	Depth d , high three bits zero.
6	1 byte	Phase f , high three bits zero.
7	1 byte	Cycle Ξ , high six bits zero.
8–15	8 bytes	Exact 64-bit word W .
16–19	4 bytes	Sequence number, modulo 2^{32} .
20–21	2 bytes	Number of valid output bits N .
22–23	2 bytes	Temperature code $H_T = \text{rnd}(100T)$, saturated with flag.
24–25	2 bytes	Payload length $L = \lceil N/8 \rceil$.
26–(25 + L)	L bytes	Slit bits, most significant bit first; unused low bits of the last byte zero.
(26 + L)–(27 + L)	2 bytes	CRC-16 check value.

Total length is $28 + L$ bytes. The zero-payload case has CRC at bytes 26–27. The CRC uses polynomial $0x1021$, initial value $0xFFFF$, no reflection, no final XOR, and processes header plus payload in transmitted order, excluding the CRC itself. These parameters define the algorithm rather than relying on an ambiguous CRC nickname.

Redundant fields in bytes 4–7 must match the corresponding fields of W ; a mismatch is a framing or integrity fault. Sequence gaps, bad CRC, malformed lengths and invalid flags are rejected. Serial timing is an implementation parameter set to the actual controller/driver limits; an abstract word layout does not specify a bus clock.

11.3 Measured record accompanying a run

[D,C] A run record contains the firmware/kernel version, LUT contents, initial word, branch sequence, clock interval, physical site map, thermal/optical calibration, sensor records, and voltage/current records. Retaining the branch sequence and measurement sequence is necessary to replay the same externally driven trajectory.

A finite deterministic digital implementation with fixed inputs eventually revisits a state and becomes periodic. The physical state need not do so because it includes continuous thermal dynamics and external measurements. Unbounded symbolic tree generation is defined by the address language $\{0, 1\}^*$; the 64-bit packet encodes the finite node currently being processed.

12 Conformance, measurements and derived predictions

12.1 Executed software checks

[P] The embedded `wantwomban_reference.py` was executed for this edition. It uses only the Python standard library and contains its own tests. The results below are execution results, not physical-device tests.

Check	Cases	Result
Packed-word extraction/reconstruction	10,000	PASS
Single-field replacement preserves other fields	10,000	PASS
Klein/radial seam applied twice	2,048	PASS
Both child transitions, range and metadata	2,000	PASS
OTAN2 inputs in each of the two profiles	4,096	PASS
Entries of the identity $BB = 4I$	16	PASS
Thermal on/off equilibria and hysteresis memory	Selected cases	PASS

The random test seed is 19900710. The source date is used here only as a reproducible seed. The source-to-PDF attachment fingerprint is recorded in Appendix C.

The integer word trace below starts from $W_0 = 0000000000100003$, uses $b_n = n \bmod 2$, keeps measured $D_n = 1$, and uses the constant reference LUT. The command column is the command issued *from* the word on that row.

n	W_n (hexadecimal)	ρ signed	θ	d	f	u_n
0	0000000000100003	0	0	0	0	0
1	0011E5FFFF0893B	1	242	1	2	0
2	26210FDEB78111C3	610	135	2	4	1
3	111D80903571999B	273	1728	3	6	1
4	306DF37CF22220CB	774	1785	4	8	1
5	68C94AB91D12A8F3	-372	1189	5	10	0

This trace tests the completed recurrence, including its quantization, rather than the earlier unparenthesized pseudocode.

12.2 Measurements that determine the physical instance

[C] Measure heater voltage and current to obtain $P(t)$. Apply a known heating step and fit C and G from temperature rise and cooling. Record reflectance on both heating and cooling to determine $T_\uparrow$, $T_\downarrow$ or the smooth material response. Measure neighbouring sites to estimate G_{jk} . Independently determine sensor delay, noise, drift and the driver cutout.

Physical initial conditions are $T(\mathbf{x}, t_0)$ and material history (a, ξ, h) ; the digital initial condition is W_0 . For every boundary or contact in the thermal model, record its prescribed temperature, measured exchange law, or insulation condition. These conditions complete the differential equations rather than serve as an exclusion of the source's proposed architecture.

12.3 Directly testable consequences

[P] With constant one-site coefficients and held command, the temperature must follow (10.3); on-state equilibrium rise is $P_{\max}/G$ and the time constant is C/G . For a chosen threshold, onset time follows the logarithmic expression in Section 10. A seam permutation with unchanged passive hardware cannot reverse heat flow between the same two temperatures.

For a measured run, the energy residual is

$$\mathcal{E} = \int_{t_0}^{t_1} P_{\text{electrical}}(t) dt - (U(t_1) - U(t_0)) - \int_{t_0}^{t_1} \dot{Q}_{\text{out}}(t) dt, \quad (12.1)$$

with any mechanical or optical input added explicitly. It should be consistent with measurement uncertainty and omitted modeled losses. Unexplained residuals call for a revised physical model or measurement account; a change of state-bit terminology is not an energy term.

For geometry, every culled descendant must satisfy the proven slit exclusion. This can be checked against an exhaustive finite-tree traversal for small depths. For a quotient-compatible material pattern, paired seam sites must agree on scalar temperature and reflectance under the chosen identification; a deliberate polarity channel is tested separately.

13 Source-to-formalization concordance

The following records the mathematical and physical completions rather than silently importing the prior generated text's claims. The source witness is [S].

Source wording or operation	Formal content in this edition
"1-bit SDF"	Retained as binary occupancy O . A signed distance is separately defined and is not represented by one bit.
"No chirality / no inside-outside"	Branch labels have no handedness. Klein nonorientability, mirror symmetry, local occupancy and a complemented parity channel are separately specified.
"OTAN2 rather than ATAN2"	OTAN2 is the source-defined occupancy-conditioned phase generator. Explicitly grouped G0 and literal left-fold Z0 have different images; both are given and tested.
"Infinite core nesting" in one word	Node addresses form an unbounded mathematical language. A packed node has finite fields; LUTs, a node collection, measurements and material memory are explicit additional state.
"0-order" C expressions	A left-fold evaluation language is defined. Actual code uses grouping and sequential assignments rather than claiming native C precedence changes.
H_4 entries " ± 1 "	The normalized tensor entries are $\pm 1/2$. The integer sign matrix is $B = 2H_4$.
Pinion operator preserves "area" or information density	The exact real operator has $ \det A_\phi = 1$ in four dimensions, not general area/length preservation. Fixed-point clipping and rounding have their own loss.
Golden-ratio operation by arbitrary shifts	The fixed coefficients 106039/65536 and 40503/65536 are specified; their products are implemented by shift/add. They approximate the irrational factors.
Chromosome-swap bit	A defined permutation of author-labelled token channels; the conflicting source decoders are resolved by an explicit truth table.
$\Gamma(t) \geq 0$ for a logarithm of a bounded product	The original log is nonpositive and singular at zeros. An explicit ε regularization and normalization yield a bounded nonnegative modulator.
"Klein eigenvalue -1 "	The quotient gluing supplies a reversed local tangent direction. The affine loop's fixed-point equation is stated separately.
Half-turn called a Klein seam	A phase reflection is required for the chosen Klein quotient. The reference uses $(1024 - \theta) \bmod 2048$; its reciprocal-radius action is a separate attached involution.
XOR all coordinate bits as numerical negation	XOR produces complement. Signed negation is extracted, computed and repacked; the symmetric radial range makes the seam involutive.
Adding 0×4 to toggle κ	Correct toggle is XOR with 0×4 ; addition can carry into the LUT.
Temperature in bits 8:5	These positions already belong to the LUT. Temperature is stored externally, with calibrated saturation and an overrange flag.
Bit 0 reused as O	Bit 0 remains part of Ξ . Geometric occupancy and optical readback are separate one-bit signals.
PSI pruning from monotonicity/-positive definiteness	The pinion map is not positive definite and can move bounds across the slit. A descendant-containing envelope proves safe pruning within an epoch.
"Conservation of informational mass" as first law	The first law is a measured balance of electrical, mechanical, thermal and stored energy. The real matrix's determinant is a different invariant.
Every shift gives heat $k_B T \ln 2$	The bound concerns logical erasure in a specified memory model. Actual heat is obtained from electrical power and thermal coupling.
All-zero coordinates as absolute thermal zero	A digital reset and absolute temperature are different variables. The thermodynamic third law is not an OTAN2 ground-state theorem.
Paint as a universal quantum threshold / complete transparency	A calibrated reflectance and hysteresis/kinetic model defines the optical response. Colour fraction and material phase fraction are distinct.
κ makes a source become a sink	κ may select a site or heater command. Cooling requires physical heat transfer; active pumping requires a powered device and its energy balance.

Source wording or operation	Formal content in this edition
Logical XOR equals optical output	XOR is retained as a logical signal. Physical reflectance is measured or derived from a specified optical model.
Conflicting byte-stream sizes	WMB1 gives one consistent versioned header, preserved word, temperature sample, payload length, bit order and CRC.

CONSOLIDATED FORMAL STATEMENT

WANTWOMBAN is defined here as a sampled-data hybrid system whose digital part compiles parity-controlled, log-polar branching states with an explicit Klein-type seam, and whose physical part converts a powered binary command into temperature-dependent optical response. Its self-reference is the causal return of measured material state to subsequent digital evolution. The named architecture, the exact mathematical statements, and the applicable physical balances are all present in one specification.

A Exact masks and bit-level repairs

Field	Shift / width	Exact 64-bit mask
χ	63 / 1	8000000000000000
ρ	52 / 11	7FF0000000000000
θ	41 / 11	000FFE0000000000
z	30 / 11	000001FFC0000000
Φ	20 / 10	000000003FF00000
d	15 / 5	00000000000F8000
f	10 / 5	0000000000007C00
ℓ	3 / 7	00000000000003F8
κ	2 / 1	0000000000000004
Ξ	0 / 2	0000000000000003

The field- θ half-turn toggles bit 51 of the complete word, with mask 0008000000000000. It does not toggle all bits in the angular field. Clearing depth uses complement mask FFFFFFFF07FFF. These operations are width-defined.

For a conditional complete-field complement controlled by bit b , use $W' = W \oplus (M \& (0 - b))$ with an unsigned width of 64. The source expression $(W \gg 63) \& M$ cannot spread a bit into a high-position mask: the shifted value is only 0 or 1, so that AND is zero when M excludes bit 0.

For a four-bit external temperature sample, clip after conversion. Arithmetic modulo 16 would turn a larger temperature into a low code and is not a valid default thermal saturation rule. Physical units belong to the calibration mapping, not the mask.

B Reference kernel

The following is the executable reference included as a PDF attachment. It implements the completed G0 controller, paired seam, constant example LUT, and a separate exact one-site thermal model. It is a reference recurrence, not a driver for unspecified hardware. Its tests cover digital identities and selected modeled thermal cases; it contains no network access, physical-device connection or external dependencies.

Run `python wantwomban_reference.py` to execute the conformance checks. A complete tree traversal supplies both branch values from the same parent and stores its stack externally. A physical feedback run obtains `optical_dark` from the measured coating; the temperature solver is available for simulation with explicitly chosen parameters.

```

1  """WANTWOMBAN v1.0: auditable reference kernel, standard library only.
2
3  The 64-bit word is geometry/control state. Temperatures, material memory,
4  LUTs, the tree stack, and measured physical parameters are separate.
5  Run this file to execute the deterministic conformance tests.
6  """
7  from math import exp, isfinite
8  from random import Random
9
10 FIELDS = {
11     "chi": (63, 1), "rho": (52, 11), "theta": (41, 11),
12     "z": (30, 11), "phi": (20, 10), "depth": (15, 5),
13     "phase": (10, 5), "lut": (3, 7), "kappa": (2, 1),
14     "cycle": (0, 2),
15 }
16 WORD_MASK = (1 << 64) - 1
17 LUT = (128,) * 128 # Example log-radius boundary, not measured paint data.
18
19 def get(w: int, name: str) -> int:
20     if not isinstance(w, int) or not 0 <= w <= WORD_MASK:
21         raise ValueError("word must be an unsigned 64-bit integer")
22     shift, width = FIELDS[name]
23     return (w >> shift) & ((1 << width) - 1)
24
25 def put(w: int, name: str, value: int) -> int:
26     get(w, name) # Validate the incoming word.
27     shift, width = FIELDS[name]
28     mask = (1 << width) - 1
29     if not isinstance(value, int) or not 0 <= value <= mask:
30         raise ValueError("field value is outside its unsigned range")
31     clear = WORD_MASK ^ (mask << shift)
32     return (w & clear) | (value << shift)
33
34 def signed(raw: int, width: int) -> int:
35     if not 0 <= raw < (1 << width):
36         raise ValueError("invalid signed-field encoding")
37     return raw - (1 << width) if raw & (1 << (width - 1)) else raw
38
39 def clip(x: int, lo: int, hi: int) -> int:
40     return min(hi, max(lo, x))
41
42 def nearest(n: int, d: int) -> int:
43     """Nearest rounding with power-of-two divisor; ties away from zero."""
44     if d <= 0 or (d & (d - 1)):
45         raise ValueError("denominator must be a positive power of two")
46     q = (abs(n) + (d >> 1)) >> (d.bit_length() - 1)
47     return -q if n < 0 else q
48
49 def constmul(x: int, coefficient: int) -> int:
50     """Exact multiplication by a nonnegative integer using shift/add."""
51     if coefficient < 0:
52         raise ValueError("coefficient must be nonnegative")
53     out = 0
54     while coefficient:
55         if coefficient & 1:
56             out += x
57         x <<= 1
58         coefficient >>= 1
59     return out

```

```

60
61 def pinion(v: tuple[int, int, int, int]) -> tuple[int, ...]:
62     x, y, z, f = v
63     a, b = x + y + z + f, x - y + z - f
64     c, d = x + y - z - f, x - y - z + f
65     return (nearest(constmul(a, 106039), 131072),
66             nearest(constmul(b, 40503), 131072),
67             nearest(c, 2), nearest(-d, 2))
68
69 def otan2(rho_raw: int, occupancy: int) -> int:
70     if occupancy not in (0, 1):
71         raise ValueError("occupancy must be one bit")
72     a = rho_raw & 2047
73     b = occupancy << 10
74     a = a ^ b
75     return a & 2047
76
77 def otan2_literal(rho_raw: int, occupancy: int) -> int:
78     if occupancy not in (0, 1):
79         raise ValueError("occupancy must be one bit")
80     return (((rho_raw & 2047) ^ occupancy) << 10) & 2047
81
82 def parity3(w: int) -> int:
83     return (w & 1) ^ ((w >> 1) & 1) ^ ((w >> 2) & 1)
84
85 def occupancy(w: int, lut: tuple[int, ...] = LUT) -> tuple[int, int]:
86     if len(lut) != 128 or any(not isinstance(x, int) for x in lut):
87         raise ValueError("LUT must contain 128 integer boundaries")
88     c = get(w, "cycle")
89     c13 = (c << 3) + (c << 2) + c
90     j = ((get(w, "theta") >> 4) + get(w, "phase") + c13) & 127
91     jitter = ((get(w, "phi") & 1) ^ get(w, "kappa"))
92     threshold = lut[j] + (jitter << 1) - 1
93     return int(signed(get(w, "rho"), 11) <= threshold), j
94
95 def seam(w: int) -> int:
96     """Shifted Klein reflection plus an independent log-radius inversion."""
97     r = signed(get(w, "rho"), 11)
98     if r == -1024:
99         raise ValueError("seam profile uses symmetric radius range")
100    w = put(w, "rho", (-r) & 2047)
101    w = put(w, "theta", (1024 - get(w, "theta")) & 2047)
102    return put(w, "kappa", get(w, "kappa") ^ 1)
103
104 def step(w: int, branch: int, optical_dark: int,
105         lut: tuple[int, ...] = LUT) -> tuple[int, int]:
106     """One node event. Returns next word and binary heater command.
107
108     optical_dark is a previously acquired measurement, never an algebraic
109     unknown. The returned command applies over the following time slot.
110     """
111     if branch not in (0, 1) or optical_dark not in (0, 1):
112         raise ValueError("branch and optical measurement must be bits")
113     g, j = occupancy(w, lut)
114     command = 1 - (g ^ get(w, "chi") ^ get(w, "kappa"))
115     p = parity3(w) ^ branch
116     f = get(w, "phi")
117     f = ((f << 1) & 1023) if p == 0 else (f >> 1)
118     r, a, z, f = pinion((signed(get(w, "rho"), 11),
119                          signed(get(w, "theta"), 11),
120                          signed(get(w, "z"), 11), signed(f, 10)))
121     r = clip(r + ((branch << 1) - 1), -1023, 1023)
122     golden = 1266 if p == 0 else -1266
123     a = (a + otan2(r & 2047, g) + golden) & 2047
124     out = put(w, "rho", r & 2047)
125     out = put(out, "theta", a)
126     out = put(out, "z", clip(z, -1024, 1023) & 2047)
127     out = put(out, "phi", f & 1023)
128     out = put(out, "lut", j)
129     out = put(out, "phase", (get(w, "phase") + 1 + optical_dark) & 31)
130     if get(w, "depth") == 31:
131         out = seam(out)
132         out = put(out, "depth", 0)
133     else:
134         out = put(out, "depth", get(w, "depth") + 1)

```

```

135     return out, command
136
137 def thermal_step(T: float, command: int, dt: float, C: float = 0.02,
138                 G: float = 0.005, Ta: float = 293.15,
139                 Pmax: float = 0.10) -> float:
140     """Exact constant-input lumped model; all temperatures are kelvin."""
141     if command not in (0, 1) or not all(map(isfinite,
142                                           (T, dt, C, G, Ta, Pmax))):
143         raise ValueError("invalid thermal arguments")
144     if min(T, C, G, Ta) <= 0 or dt < 0 or Pmax < 0:
145         raise ValueError("nonphysical lumped-model parameters")
146     target = Ta + Pmax * command / G
147     return target + (T - target) * exp(-G * dt / C)
148
149 def dark_bit(T: float, previous: int, low: float = 301.15,
150             high: float = 304.15) -> int:
151     if previous not in (0, 1) or not low < high or not isfinite(T):
152         raise ValueError("invalid hysteresis arguments")
153     return 0 if T >= high else (1 if T <= low else previous)
154
155 def self_test() -> dict[str, int]:
156     rng = Random(19900710)
157     masks = [(1 << width) - 1] << shift for shift, width in FIELDS.values()
158     assert sum(masks) == WORD_MASK
159     assert all(not (a & b) for i, a in enumerate(masks) for b in masks[i+1:])
160     for _ in range(10000):
161         w = rng.getrandbits(64)
162         out = 0
163         for name in FIELDS:
164             out = put(out, name, get(w, name))
165         assert out == w
166         name = rng.choice(tuple(FIELDS))
167         s, n = FIELDS[name]
168         out = put(w, name, rng.randrange(1 << n))
169         mask = ((1 << n) - 1) << s
170         assert ((out ^ w) & (WORD_MASK ^ mask)) == 0
171     for o in (0, 1):
172         assert len({otan2(r, o) for r in range(2048)}) == 2048
173         assert {otan2_literal(r, o) for r in range(2048)} == {0, 1024}
174     for a in range(2048):
175         w = put(put(put(0, "theta", a), "rho", 77), "kappa", a & 1)
176         assert seam(seam(w)) == w
177     B = ((1,1,1,1), (1,-1,1,-1), (1,1,-1,-1), (1,-1,-1,1))
178     for i in range(4):
179         for j in range(4):
180             assert sum(B[i][k]*B[k][j] for k in range(4)) == (4 if i==j else 0)
181     for _ in range(1000):
182         w = rng.getrandbits(64)
183         for b in (0, 1):
184             out, u = step(w, b, rng.randrange(2))
185             assert 0 <= out <= WORD_MASK and u in (0, 1)
186             assert get(out, "chi") == get(w, "chi")
187             assert get(out, "cycle") == get(w, "cycle")
188             assert get(out, "depth") == (get(w, "depth") + 1) % 32
189     assert abs(thermal_step(313.15, 1, 1) - 313.15) < 1e-10
190     assert abs(thermal_step(293.15, 0, 1) - 293.15) < 1e-10
191     assert dark_bit(302, 0) == 0 and dark_bit(302, 1) == 1
192     return {"word_roundtrips": 10000, "field_isolations": 10000,
193           "seam_involutions": 2048, "transition_cases": 2000,
194           "OTAN2_inputs_per_profile": 4096, "matrix_entries": 16}
195
196 if __name__ == "__main__":
197     print("PASS", self_test())

```

C Source witness and edition record

Source file: Pasted markdown.md.

Exact size: 61,608 bytes.

SHA-256:

1869efeab782774dd7af09cc2ca9e4dfd7f0

317acf1549a375bbb4f8cb53d136

The source is embedded unchanged. It records the originating prompts and the earlier generated explanations, including their displayed links and pseudocode. Those displayed links are part of the witness; the verified references used for this edition are listed separately below. The reference-kernel attachment is the code printed in Appendix B.

The attribution printed on the cover is the designation supplied in the request: *Tom Klootwijk, NL200678942, 10-07-1990*. The identifier and date are reproduced as authorship metadata, not used to infer an external identity or material property. The model construction, disambiguations, calculations and software tests in this edition are provided as additions to that supplied framework.

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